

CHEMICAL THERMODYNAMICS AND THERMOCHEMISTRY

The Thermodynamic Properties of S-Lactic Acid

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Abstract—The enthalpies of combustion and formation of S-lactic acid at 298.15 K, $\Delta_c H_m^\circ(\text{cr.}) = -1337.9 \pm 0.8$ and $\Delta_f H_m^\circ(\text{cr.}) = -700.1 \pm 0.9$ kJ/mol, were determined by calorimetry. The temperature dependence of acid vapor pressure was studied by the transpiration method, and the enthalpy of its vaporization was obtained, $\Delta_{\text{vap}} H^\circ(298.15 \text{ K}) = 69.1 \pm 1.0$ kJ/mol. The temperature and enthalpy of fusion, T_m (330.4 K) and $\Delta_m H^\circ(298.15 \text{ K}) = 14.7 \pm 0.2$ kJ/mol, were determined by differential scanning calorimetry. The enthalpy of formation of the acid in the gas phase was obtained. Ab initio methods were used to perform a conformational analysis of the acid, calculate fundamental vibration frequencies, moments of inertia, and total and relative energies of the stablest conformers. Thermodynamic properties were calculated in the ideal gas state over the temperature range 0–1500 K. A thermodynamic analysis of mutual transformation processes (the formation of SS- and RS(meso)-lactides from S-lactic acid and the racemization of these lactides) and the formation of poly-(RS)-lactide from S-lactic acid and SS- and RS(meso)-lactides was performed.

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INTRODUCTION

S(+)-Lactic acid (2-hydroxypropanoic acid) is extensively used in mordant dyeing and in food, leather, cosmetic, pharmaceutical, and other industries. It is of special importance for polymer industry as a monomer for the preparation of a biodegradable polymer, polylactide, which is extensively used in medicine and the production of film materials.

At the same time, the thermodynamic properties of lactic acid have been studied poorly and require refinement. This is primarily related to the complexity of the production of pure lactic acid, which experiences partial conversion to various anhydro derivatives when allowed to stand in aqueous solutions even at room temperature [1]. For instance, the temperature of fusion varies from 298–299 [2, 3] to 325–327 K [4–6] according to various authors. The enthalpies of combustion and formation of a sample prepared in a series of fractional crystallizations of a commercial syrup were measured in 1959 [7], but the absence of data on sample purity and the method for calculating measurement errors decreases the reliability of the result. In [8], the temperature dependence of heat capacity was measured over the temperature range 84–311 K for optically pure R- and S-lactic acid samples containing 99.7% major component; the entropy of crystal compounds at 298.15 K was determined. In the same work, the enthalpy of fusion of R-lactic acid was estimated (16.86 kJ/mol). However, according to

[1], temperature measurements by a thermocouple that was not calibrated according to the generally accepted scale and the formation of acid anhydro derivatives during melting do not allow the data obtained in [8] to be considered reliable.

In this work, we experimentally obtained the enthalpies of combustion and formation of pure S-lactic acid, measured the enthalpy and temperature of fusion, the temperature dependence of vapor pressure, and the enthalpy of vaporization, performed quantum-chemical calculations, and used the molecular and spectral data to calculate the thermodynamic properties of the compound in the ideal gas state over the temperature range 0–1500 K. We performed a thermodynamic analysis of mutual conversion of SS- and RS-lactides and S-lactic acid in the formation of poly-(RS)-lactide.

EXPERIMENTAL

Sample preparation. Commercial S-lactic acid (Alfa Aesar, dehydrated, >98%) was additionally purified by sublimation in a vacuum. Sample purity was determined by capillary chromatography, the sample contained no less than 99.8% major component. The amount of moisture in the sample (417 ppm) was determined by conductometric titration.

Combustion calorimetry. The heat of combustion of crystalline S-lactic acid was measured in a modified V-08MA isothermic-shell calorimeter with a sta-

Table 1. Calorimetric data on the combustion of S-lactic acid ($T = 298.15$ K, $p^\circ = 0.1$ MPa)

m , g	m' , g	m'' , g	ΔT_c , K	$\frac{-(\varepsilon_{\text{calor}})}{\times (-\Delta T_c)}$, J	$\frac{-(\varepsilon_{\text{cont}})}{\times (-\Delta T_c)}$, J	$\frac{\Delta U_{\text{decomp}}}{(\text{HNO}_3)}$, J	ΔU_{corr} , J	$-m'\Delta_c u'$, J	$-m''\Delta_c u''$, J	$-\Delta_c u^\circ$ (cr), J/g
0.456978	0.000989	0.290086	1.36882	20269.8	25.21	36.13	9.73	16.76	13446.79	14848.8
0.688645	0.001131	0.295015	1.61779	23956.6	30.89	38.82	13.16	19.16	13675.27	14871.3
0.728367	0.001163	0.232339	1.46056	21628.3	27.66	34.94	12.66	19.71	10769.96	14853.4
0.592246	0.001021	0.227391	1.30830	19373.5	24.16	33.15	10.64	17.30	10540.60	14852.0
0.522830	0.001106	0.248765	1.30600	19339.5	24.05	33.75	9.99	18.74	11531.38	14860.9
1.004265	0.000988	0.153259	1.48979	22061.0	28.68	34.64	15.31	16.74	7104.24	14855.4
0.757305	0.000808	0.161265	1.26712	18763.7	23.65	30.16	11.88	13.69	7475.36	14863.6

Note: m , m' , and m'' are the weights of the substance, cotton, and polyethylene, respectively; the other symbols are explained in [11]; $\langle -\Delta_c u^\circ \rangle = 14857.9 \pm 3.0$ J/g.

Table 2. S-Lactic acid vapor pressures and enthalpies of vaporization (kJ/mol) measured by the transpiration method

T , K	m , mg	$V(\text{N}_2)$, dm ³	p , Pa	$p_{\text{expt}} - p_{\text{calcd}}$, Pa	$\Delta_{\text{vap}} H^\circ$
333.8	18.9	19.3	27.97	-0.46	66.89
338.8	8.50	6.00	39.86	-0.68	66.58
343.4	22.8	10.9	58.47	2.67	66.29
348.4	28.0	9.73	79.73	2.18	65.98
353.4	13.8	3.67	104.0	-3.02	65.67
358.4	20.3	3.93	142.1	-3.90	65.36
363.4	33.5	4.60	200.1	2.72	65.05
368.4	27.0	2.80	264.5	0.36	64.74

Note: The mean value is $\Delta_{\text{vap}} H^\circ(298.15 \text{ K}) = 69.1 \pm 1.0$ kJ/mol; $\Delta_{\text{vap}} H^\circ = 65.9 \pm 1.0$ kJ/mol at the middle of the temperature range of measurements (351.1 K); the rate of carrier gas $W(\text{N}_2) = 4.0$ dm³/h.

tionary bomb. The procedure for measurements and special features of performing calorimetric experiments were described in [9, 10]. The energy equivalent of the calorimeter (ε) was determined by measurements of the heat of combustion of NIST SMR 39j standard benzoic acid. According to the results of 12 experiments, $\varepsilon = 14808.2 \pm 0.9$ J/K. We used polyethylene with a known enthalpy of combustion (-46354.5 ± 3.7 J/g) as an auxiliary substance. The results of calorimetric measurements are listed in Table 1.

The determination of vapor pressures and the enthalpy of vaporization. The saturated vapor pressure and the enthalpy of vaporization of S-lactic acid were determined at 334–368 K by the transpiration method [12], the weight of the sample was measured gravimetrically. The accuracy of weighing was 1×10^{-4} g. The temperature dependence of vapor pressure had the form

$$\ln p [\text{Pa}] = \frac{297.2}{R} - \frac{87581.1}{R(T)} - \frac{62.0}{R} \ln \left(\frac{T}{298.15} \right).$$

The experimental results are listed in Table 2.

Differential scanning calorimetry (DSC). The temperature and enthalpy of fusion of the compound were

measured using a PerkinElmer DSC-2C differential scanning calorimeter. The procedure for measurements was described in [12]. The calorimeter was calibrated against high-purity indium ($T_m = 429.8$ K, $\Delta_m H^\circ = 28.5$ J/g). The temperature of fusion of lactic acid was found to be 330.4 K; at this temperature, $\Delta_m H^\circ = 16.05$ kJ/mol. Measurement errors were ± 0.5 K for the temperature of fusion and ± 0.2 kJ/mol for the enthalpy of fusion. The extrapolation of the enthalpies of vaporization and fusion to standard conditions was performed taking into account the difference between the heat capacities of the acid in the corresponding states of aggregation. The $\Delta_1^g C_p$ and $\Delta_{cr}^g C_p$ values were found as recommended in [13], they were 62.0 and 19.1 J/(mol K), respectively. The thermodynamic properties of S-lactic acid (298.15 K) are summarized in Table 3.

CALCULATIONS ACCORDING TO THE MOLECULAR AND SPECTRAL DATA

A conformational analysis of lactic acid was performed at the HF/6-31G(d) level using the GAUS-

Table 3. Enthalpy characteristics of S-lactic acid at 298.15 K, kJ/mol

$\Delta_c H_m^\circ$ (cr)	$\Delta_f H_m^\circ$ (cr)	$\Delta_m H^\circ$	$\Delta_{\text{vap}} H^\circ$	$\Delta_{\text{sub}} H^\circ$	$\Delta_f H_m^\circ$ (g)
-1338.4 ± 0.6	-699.7 ± 0.7	14.7 ± 0.2	69.1 ± 1.0	83.8 ± 1.0	-615.9 ± 1.3
-1344.0 ± 0.9 [7]	-694.1 ± 1.0 [7]	—	—	—	$-610.3 \pm 1.4^*$

* Found from the $\Delta_f H_m^\circ$ (cr) [7] and $\Delta_{\text{sub}} H^\circ$ values.

SIAN 03 program, revision C02 [14]. We calculated the energies of all the conformers formed when molecule fragments were sequentially rotated about two bonds (the C—O bond with the alcoholic hydroxyl and the C—C bond with the acid group) through 360° in steps of 10° . An analysis of the internal rotation potential surface obtained this way showed that there were five stablest conformers. The optimization of their geometries and calculations of vibrational and energy characteristics were performed at the B3LYP/6-311++G(3df, 3pd) level [15]. The conformational distribution obtained (figure) was in agreement with the results of earlier experimental and theoretical studies [16–19].

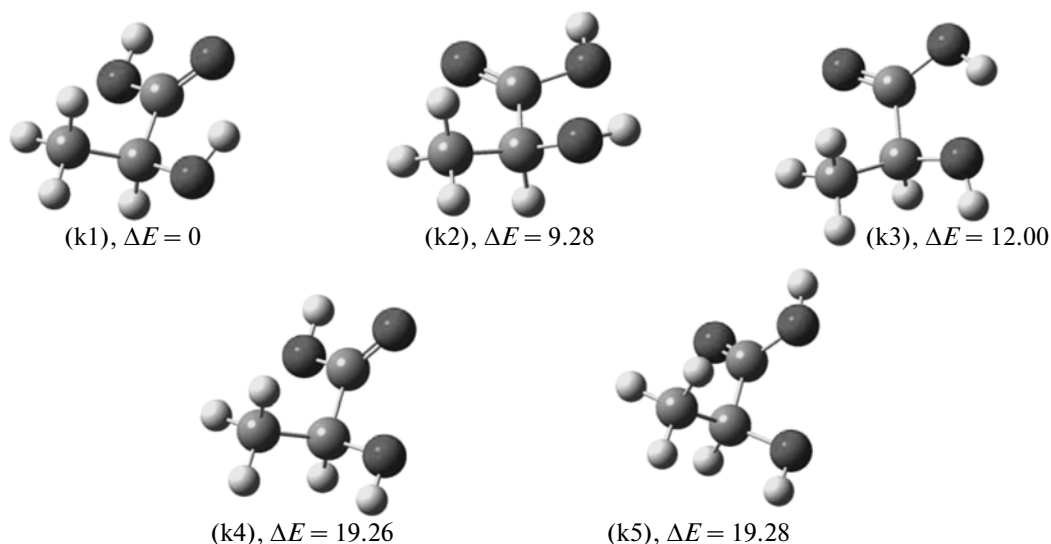
A comparison of the experimental and calculated spectra interpreted by us for conformations k1–k5 taking into account scale factors (Table 4 contains the spectrum of the k1 stereoisomeric form) shows that, as with glycolic acid, the stablest lactic acid conformers (k1–k3) are stabilized by intramolecular H-bonds. For instance, the presence of an H-bond between alcoholic hydroxyl hydrogen and acid group carbonyl in k1 is substantiated by a decrease in the stretching vibration frequencies of the alcoholic hydroxyl $\nu_{\text{OH}_{\text{al}}}$ (by $\sim 100 \text{ cm}^{-1}$) and carbonyl group $\nu_{\text{C=O}}$ (by $\sim 50 \text{ cm}^{-1}$) compared with the other conformers. A decrease in $\nu_{\text{OH}_{\text{al}}}$ by $\sim 25 \text{ cm}^{-1}$ is indicative of interaction between

two hydroxyls in k2. Conformer k3 contains an H-bond between the acid and alcoholic OH groups, as follows from a substantial (by $\sim 80 \text{ cm}^{-1}$) decrease in the frequency of acid hydroxyl vibrations $\nu_{\text{OH}_{\text{ac}}}$. In addition, an increase in the frequency of bending vibrations of the alcoholic (by $\sim 35 \text{ cm}^{-1}$ in k1–k3) and acid (by $\sim 35, 40$, and 65 cm^{-1} in k1, k2, and k3, respectively) COH groups substantiates the participation of these groups in the interactions specified.

The energy of H-bonds between the alcoholic hydroxyl hydrogen and C=O group oxygen atoms was estimated at $\sim -19 \text{ kJ/mol}$ from the energy difference between conformers k1 and k4.

The determination of the thermodynamic functions of S-lactic acid from the molecular and spectral data was performed for the stablest S-lactic acid conformer k1 following the procedure described in [20, 21]. In calculations of the vibrational component, the wave numbers missing in the experimental spectra [19, 22, 23] were taken from the calculated spectrum of k1 (the scale factors were 1.0013 and 0.9554 over the frequency ranges 0–2900 and 2990–4000 cm^{-1} , respectively) (Table 4). The moments of inertia of the acid molecule were obtained in microwave measurements [24], $I_A I_B I_C = 15548.89 \times 10^{-117} \text{ g}^3 \text{ cm}^6$.

The mixing functions for five stereoisomeric forms were found following the procedure described in [20].



Structure and relative energies (ΔE , kJ/mol) of the stablest S-lactic acid conformers.

Table 4. Comparison of the experimental and calculated spectra of the stablest S-lactic acid conformer (k1), cm^{-1}

Experimental frequencies						Calculated	Assignment
[19]		[22]		[23]			
IR(Ar)	IR(Xe)	IR	Raman	IR	Raman		
3561	3546	—	—	—	3500	3583	$\nu_{\text{OH}_{\text{ac}}}$
3559	3531	—	—	—	—	3572	$\nu_{\text{OH}_{\text{al}}}$
3004	2992	—	2990	2990	2999	2986	$\nu_{\text{CH}_3}^{\text{as}}$
2998	2982	—	2945	2943	2948	2978	$\nu_{\text{CH}_3}^{\text{as}}, \nu_{\text{CH}}$
—	—	—	2910	2833	2900	2908	$\nu_{\text{CH}_3}^{\text{s}}, \nu_{\text{CH}}$
2949	2938	—	2890	2010	2730	3005	ν_{CH}
1769	1762	1725	1725	1732	1830	1804	$\nu_{\text{C=O}}$
1464	1460	1475	1475	—	1550	1503	$\delta_{\text{CH}_3}^{\text{as}}$
1459	1455	1455	1455	1456	1456	1496	$\delta_{\text{CH}_3}^{\text{as}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CH}}, \nu_{\text{CC}}$
1376	1365	1420	1420	—	—	1437	$\delta_{\text{CH}_3}^{\text{as}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CH}}, \delta_{\text{COH}_{\text{ac}}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
1381	1374	1380	1380	1376	—	1409	$\delta_{\text{CH}_3}^{\text{s}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CH}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
—	—	1335	1330	—	—	1352	$\delta_{\text{CH}}, \delta_{\text{CH}_3}^{\text{wagg}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CCC}}, \delta_{\text{CCO}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
1281	1276	1285	—	—	—	1339	$\delta_{\text{COH}_{\text{ac}}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CCO}}, \delta_{\text{CCC}}, \delta_{\text{CH}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
1253	1249	1240	1245	—	—	1277	$\delta_{\text{COH}_{\text{al}}}, \delta_{\text{COH}_{\text{ac}}}, \delta_{\text{CH}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
—	—	—	—	1220	1230	1190	$\gamma_{\text{CCC}}, \delta_{\text{CCO}}, \delta_{\text{COH}_{\text{ac}}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{CH}_3}^{\text{wagg}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
1119	1117	1130	1135	—	1190	1145	$\delta_{\text{COH}_{\text{ac}}}, \nu_{\text{CO}}, \nu_{\text{CC}}, \gamma_{\text{CCC}}$
1162	1173	1190	1190	1128	1140	1107	$\delta_{\text{CCC}}, \delta_{\text{CH}_3}^{\text{wagg}}, \nu_{\text{CC}}, \nu_{\text{CO}}$
1094	1094	1050	1050	1047	1030	1050	$\delta_{\text{CCC}}, \nu_{\text{CC}}, \delta_{\text{CH}_3}^{\text{wagg}}, \delta_{\text{COH}_{\text{al}}}, \delta_{\text{COH}_{\text{ac}}}, \nu_{\text{CO}}$
1040	1040	930	930	923	925	934	$\delta_{\text{CH}_3}^{\text{rock}}, \nu_{\text{CC}}, \delta_{\text{CCC}}, \delta_{\text{COH}_{\text{ac}}}$
803	802	—	830	823	827	810	$\delta_{\text{CCC}}, \delta_{\text{COH}_{\text{ac}}}$
734	727	—	750	747	747	744	$\delta_{\text{CCC}}, \delta_{\text{OCO}}, \delta_{\text{COH}_{\text{ac}}}$
634	619	—	650	642	640	640	$\delta_{\text{COH}_{\text{ac}}}, \delta_{\text{CCO}}, \delta_{\text{CCC}}$
560	556	—	540	637	—	581	$\gamma_{\text{COH}_{\text{ac}}}, \gamma_{\text{CCC}}$
492	—	—	425	546	—	499	$\delta_{\text{CCC}}, \delta_{\text{CCO}}$
—	—	—	—	541	—	410	$\gamma_{\text{COH}_{\text{al}}}, \gamma_{\text{CCC}}$
—	—	—	370	—	—	354	δ_{CCC}
—	—	—	—	—	—	307	δ_{CCC}
—	—	—	275	—	—	247	τ_{sk}
—	—	—	—	—	—	230	τ_{sk}
—	—	—	—	—	—	56	τ_{sk}

Note: Frequencies used in property calculations are typeset in boldface; calculated frequencies are scaled, scale factors 1.0013 and 0.9554 over frequency ranges 0–2900 and 2990–4000 cm^{-1} , respectively.

Table 5. Thermodynamic properties of S- and (±)-lactic acids, ideal gas

T, K	C_p°	S_m°	$\frac{H_T^\circ - H_0^\circ}{T}$	Φ_m°	$\Delta_f H_m^\circ$	$\Delta_f G_m^\circ$	C_p°	S_m°	$\frac{H_T^\circ - H_0^\circ}{T}$	Φ_m°
S-Lactic acid						(±)-Lactic acid				
0	0.00	0.00	0.00	0.00	−594.60	−594.60	0.00	0.00	0.00	0.00
100	52.71	253.51	41.52	212.00	−603.98	−572.83	52.73	253.52	41.52	212.00
150	67.19	277.62	47.66	229.96	−607.56	−556.45	67.51	277.66	47.70	229.97
200	81.50	298.91	54.33	244.58	−610.68	−538.93	82.78	299.17	54.55	244.62
273.16	102.42	327.41	64.41	263.00	−614.70	−512.00	105.70	328.35	65.18	263.18
298.15	109.53	336.68	67.89	268.79	−615.90	−502.55	113.41	337.94	68.90	269.04
300	110.05	337.36	68.15	269.21	−615.98	−501.84	113.97	338.64	69.17	269.47
400	136.81	372.77	82.05	290.72	−619.76	−463.16	141.83	375.38	83.97	291.41
500	159.10	405.78	95.31	310.47	−622.02	−423.66	163.51	409.48	97.82	311.66
600	176.72	436.41	107.47	328.94	−623.08	−383.77	179.89	440.80	110.19	330.61
700	190.66	464.74	118.40	346.35	−623.17	−343.70	192.70	469.53	121.10	348.43
800	201.96	490.96	128.16	362.81	−622.50	−303.41	203.15	495.97	130.72	365.24
900	211.35	515.31	136.90	378.41	−621.20	−263.51	211.98	520.42	139.27	381.14
1000	219.30	538.00	144.75	393.25	−619.40	−223.52	219.56	543.15	146.93	396.22
1100	226.11	559.23	151.84	407.39	−617.17	−183.64	226.14	564.39	153.84	410.56
1200	231.99	579.16	158.28	420.88	−614.60	−143.90	231.88	584.32	160.11	424.21
1300	237.09	597.94	164.15	433.78	−611.75	−104.28	236.90	603.09	165.83	437.26
1400	241.54	615.67	169.53	446.15	−608.69	−64.86	241.30	620.81	171.06	449.74
1500	245.42	632.47	174.46	458.01	−605.45	−25.57	245.16	637.59	175.88	461.71

Note: $\Delta_f H_m^\circ$ and $\Delta_f G_m^\circ$ are in kJ/mol, the other functions are in J/(mol K).

The compositions of equilibrium mixtures of conformers at different temperatures were calculated taking into account their relative energies on the assumption of the equality of their entropies. The enthalpies and Gibbs energies were extrapolated to a broad temperature interval using the thermodynamic functions of the elements from [25]. The thermodynamic functions of S-lactic acid were used to calculate the thermodynamic functions of racemic (±)-lactic acid in the ideal gas state. The calculations were performed taking into account contributions of mixing of five pairs of mirror-isomeric forms following the procedure described in [20]. The results are listed in Table 5.

RESULTS AND DISCUSSION

The enthalpy of vaporization of S-lactic acid was measured for the first time. Reliable values of the temperature and enthalpy of fusion were obtained. The measured enthalpy of formation of S-lactic acid, $\Delta_f H_m^\circ(\text{cr.}, 298.15 \text{ K}) = -699.7 \pm 0.7 \text{ kJ/mol}$, differs from that reported in [7] ($-694.1 \pm 1.0 \text{ kJ/mol}$) by 5.6 kJ/mol. We used our data to determine the enthalpies of sublimation and formation of the compound in the gas phase (Table 3).

The reliability of the values obtained was additionally estimated by applying the G3MP2 composite method to determine $\Delta_f H_m^\circ(\text{g})$ of propionic and S-lactic acids with the use of the standard thermodynamic procedures “from the atomization reaction” and “from bond separation reactions.” Propionic acid was used as a test substance with a similar structure and reliable thermodynamic characteristics at 298.15 K, $\Delta_f H_m^\circ(\text{l}) = -510.8 \pm 1.0 \text{ kJ/mol}$ [26] and $\Delta_{\text{vap}} H^\circ = 55.0 \pm 2.0 \text{ kJ/mol}$ [27], whence $\Delta_f H_m^\circ(\text{g}) = -455.8 \pm 2.2 \text{ kJ/mol}$. The calculation results are listed in Table 6. Close agreement between the calculated and experimental $\Delta_f H_m^\circ$ values for both acids allows the enthalpy of formation obtained in this work to be characterized as reliable.

The absolute entropy of S-lactic acid at 298.15 K in the gas state obtained from the molecular and spectral data (336.68 J/(mol K)) is larger by 5.11 J/(mol K) than that measured for the crystalline state in [8] and recalculated for the gas phase using the enthalpies of vaporization and fusion and vapor pressures obtained in this work ($S_m^\circ(\text{g}, 298.15 \text{ K}) = 331.57 \text{ J/(mol K)}$). However, because of a large extrapolation interval to

Table 6. Quantum-chemical data on $\Delta_f H_m^\circ$ (g) of propionic and S-lactic acids at 298.15 K, kJ/mol

Reaction	$-\Delta_f H_m^\circ$ (g)
Propionic acid	
$C_2H_5COOH + 5C_2H_6 = 2C_2H_5OH + 3C_3H_8$	455.5
$C_2H_5COOH + C_2H_6 = CH_3COOH + C_3H_8$	455.0
$C_2H_5COOH + 9C_2H_6 = 2H_2O + 7C_3H_8$	455.2
From atomization reaction	450.7
Mean	454.1
S-Lactic acid	
$C_2H_4(OH)COOH + 6C_2H_6 = 3C_2H_5OH + 3C_3H_8$	618.5
$C_2H_4(OH)COOH + 12C_2H_6 = 3H_2O + 9C_3H_8$	618.1
$C_2H_4(OH)COOH + 3C_3H_8 = CH_3COCH_3 + C_2H_5COOH + 3C_2H_6$	618.1
From atomization reaction	611.6
Mean	616.6

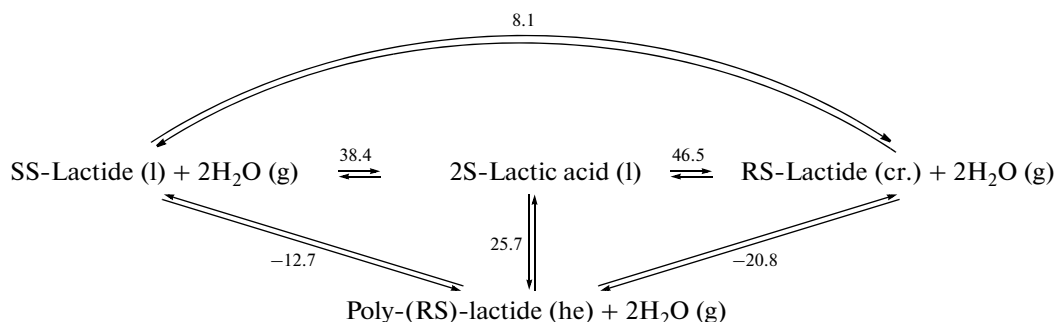
Note: The enthalpies of formation of reaction participants were taken from [28].

0 K (from 90 K) and incomplete purification of S-lactic acid (99.7%), the entropy in the condensed state obtained in [8] is likely not very reliable and cannot be used as a reference for bringing the calculation results in agreement with it.

The thermodynamic data (the enthalpies of formation and absolute entropies of compounds in various condensed states) on amorphous poly-(RS)-lactide

[29] and RS-(*meso*) and SS-lactides [30, 31] and the results obtained in this work for S-lactic acid (Tables 2, 3, 5) were used to calculate the thermodynamic characteristics of reactions and equilibrium constants of the formation of lactides from S-lactic acid, their racemization, and the preparation of poly-(RS)-lactide from the compounds specified (per polymer chain recurring unit). The racemization of lactic acid above 130–140°C [32, 33] and optically active lactide above $t \geq 180^\circ\text{C}$ [34] is possible under polymer preparation conditions. It follows that polymers with different contents and sequences of stereoisomeric units can form from optically pure initial reagents.

The preparation of RS- and SS-lactides and poly-(RS)-lactide from S-lactic acid is characterized by small positive Gibbs energies of the same order of magnitude. For this reason, the polycondensation of lactic acid gives lactides in addition to the polymer. To shift equilibrium toward the formation of the products specified, temperature in the reaction zone should be increased and water released should be removed. The polymerization of RS- and SS-lactides with ring opening and polymer formation is allowed thermodynamically over the temperature interval under consideration (100–390 K). The equilibrium constants of mutual conversion of optically active lactide and its *meso*-form are close to zero and change insignificantly over the temperature range studied (100–450 K), and their equilibrium mixtures contain stereoisomers in comparable amounts. A scheme of equilibrium transformations of the compounds at 390 K with the corresponding $\Delta_r G_m^\circ$ values (kJ/mol) is as follows:



The thermodynamic properties of $H_2O(g)$ were found by the extrapolation of the corresponding values from [38]; substance physical states: cr, crystalline; he, highly elastic; l, liquid; and g, gaseous.

The results of our analysis substantiate the conditions of the occurrence of processes in the synthesis of the polymer [35–37].

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